

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2339

Arcanum Peaking 100 kV
September 28, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2339. The Interconnection Customer proposes a 106.6 MW battery energy storage facility interconnecting at the Arcanum Peaking 100 kV substation in the Ohio area. The request was received on August 08, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 346 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$37,488,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Snyder / Eastmont 100 kV	W Asheville / Websterville 100 kV L	145.7	166	41.7
Fiddymment / Miami 100 kV line	Baraga Vlg / Queen Crk 500 kV line	129.6	184	5.3
West Asheville Substation / Websterville 100 kV Line	Snyder / Eastmont 100 kV	125.6	194	31.3
Newcomb / Arcanum Peaking 100 kV	Wissota / Berry St 138 kV	113.4	129	100.0
Powerline Elkhorn Substation / Thousand Springs 100 kV	Rancho Santa Fe / Snyder 100 kV	112.2	309	63.3
Rancho Santa Fe / Blount 100 kV	Powerline Elkhorn / Websterville 100 kV	111.9	311	66.2
Snyder / Rancho Santa Fe 100 kV	Powerline Elkhorn / Websterville 100 kV	111.7	320	66.2
Newcomb / Blount 100 kV	Wissota / Berry St 138 kV	109.4	126	100.0
Websterville / Powerline Elkhorn 100 kV	Snyder Substation / Rancho Sata Fe 100 kV	109.3	337	63.3
Fiddymment 100/1kV XFMR	Powerline Elkhorn Substation / Websterville Substation 100 kV	107.0	162	6.3
Aep Medford / Fairway Tap 100 kV	Headwaters Wind Farm li Llc / Eaton 100 kV	101.0	160	15.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Eastmont / Snyder 100 kV L	Rebuild line with higher-capacity conductor	\$3,481,000	35% (\$1,204,000)
Fiddymment / Miami 100 kV line	Rebuild line with higher-capacity conductor	\$3,582,000	100% (\$3,582,000)
Websterville / West Asheville 100 kV L	Rebuild line with higher-capacity conductor	\$28,255,000	37% (\$10,334,000)
Newcomb / Arcanum Peaking 100 kV	Rebuild line with higher-capacity conductor	\$2,103,000	100% (\$2,103,000)
Powerline Elkhorn Substation / Thousand Springs 100 kV	Rebuild line with higher-capacity conductor	\$2,257,000	64% (\$1,437,000)
Blount / Rancho Santa Fe 100 kV line	Rebuild line with higher-capacity conductor	\$3,038,000	62% (\$1,899,000)
Snyder / Rancho Santa Fe 100 kV	Rebuild line with higher-capacity conductor	\$4,899,000	62% (\$3,062,000)
Newcomb / Blount 100 kV	Reconductor line	\$4,808,000	100% (\$4,808,000)
Powerline Elkhorn / Websterville 100 kV	Reconductor line	\$4,985,000	64% (\$3,174,000)
Fiddymment 100/1 kV XFMR	Replace transformer with higher-rated unit	\$3,482,000	100% (\$3,482,000)
Aep Medford / Fairway Tap 100 kV	Reconductor line	\$6,604,000	36% (\$2,403,000)
	Total allocated to project		\$37,488,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.